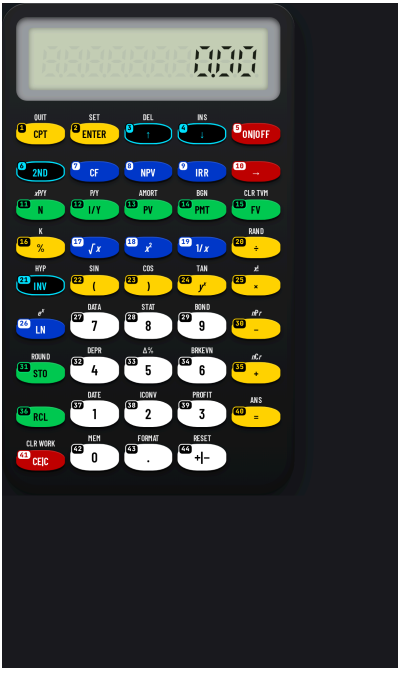
U-45 CALCULATORS

User Manual

Calculator #9 Financial Business Analyst

Financial



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#9 — Financial Business Analyst** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example [#1 CPT]. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

financial AOS with worksheets (= completes; TVM keys store and CPT computes).

Chapters

1. Getting started
2. Features
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4. Key-by-key reference
5. Worked examples
6. Tricks & hacks
7. Keyboard shortcuts
8. Common mistakes
9. Practice drills
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14. Appendix E — Quick-reference card

1 · Getting started

Open U-45 CALCULATORS, choose **#9** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#27, #28, #29, #32, #33, #34, #37, #38, #39, #42, #43, #44
op	Operators & execute	#1, #2, #16, #20, #22, #23, #24, #25, #30, #35, #40
fn	Functions	#7, #8, #9, #17, #18, #19, #26
mem	Memory, variables & registers	#11, #12, #13, #14, #15, #31, #36
clr	Clear, delete & power	#5, #10, #41
mod	Shift, modes, menus & navigation	#3, #4, #6, #21

2 · Features

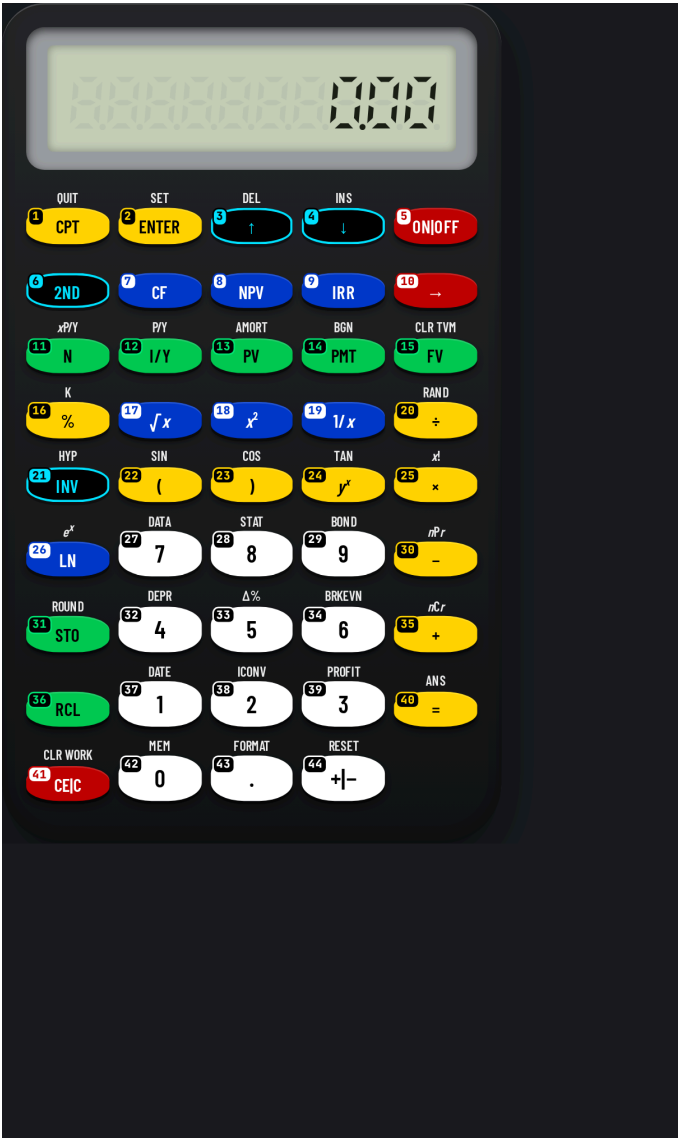
54 features: 37 live on the keys, 9 app/UI features, 8 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **TVM: N, I/Y, PV, PMT, FV with CPT for the unknown**
2. LIVE **P/Y and C/Y payments / compounding per year**
3. LIVE **BGN / END annuity mode**
4. LIVE **AMORT schedule: principal, interest, balance for P1–P2**
5. LIVE **Cash flows CF0, Cnn with frequencies; NPV and IRR**
6. LIVE **DATA / STAT worksheet with LIN, LN, EXP, PWR regressions**
7. LIVE **BOND worksheet**
8. LIVE **DEPR straight-line**
9. LIVE **Δ% percent change**
10. LIVE **BRKEVN break-even**
11. LIVE **PROFIT cost / sell / margin**
12. LIVE **ICONV nominal ↔ effective rate**
13. LIVE **DATE days between dates**
14. LIVE **MEM ten memories**
15. LIVE **FORMAT decimals and date format**
16. LIVE **CLR TVM / CLR WORK**
17. LIVE **Square root √**
18. LIVE **Square x^2**
19. LIVE **Reciprocal x^{-1} (1/x)**
20. LIVE **Power x^y / ^ with right-to-left chaining (2^3^2 = 512)**

21. LIVE **Natural logarithm ln**
22. LIVE **Exponential e^x**
23. LIVE **Trigonometry sin / cos / tan honouring the current angle unit**
24. LIVE **Hyperbolic sinh / cosh / tanh and their inverses via hyp**
25. LIVE **Factorial x! (exact up to 69!)**
26. LIVE **Permutations nPr**
27. LIVE **Combinations nCr**
28. LIVE **Random number Ran# (0 ≤ r < 1)**
29. LIVE **Rnd — round the value to the displayed precision**
30. LIVE **Percent key (x %, add-on and discount chains)**
31. LIVE **Sign change (−) / +/− / CHS**
32. LIVE **Ans recall of the last result; operators auto-insert Ans**
33. LIVE **Store a value into memory / a variable (STO, sto►, →, Min)**
34. LIVE **Recall memory / variables (RCL, MR, recall)**
35. LIVE **Backspace / DEL editing of the current entry**
36. LIVE **INS insert mode for editing in the middle of an expression**
37. LIVE **Constant calculations (K) on repeated operators**
38. APP/UI **Every key numbered 1…N in reading order — matches the key map in the user manual**
39. APP/UI **High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle**
40. APP/UI **Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)**

41. APP/UI Show / hide key numbers with one button (remembered between sessions)
42. APP/UI Copy the current display text to the clipboard
43. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
44. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
45. APP/UI Reference photo viewer and tool-tip name on every key
46. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)
47. PLANNED MIRR / NFV / payback
48. PLANNED SYD, DB, DDB depreciation
49. PLANNED Bond yield to call
50. PLANNED Loan-comparison side-by-side
51. PLANNED Amortisation table export (CSV)
52. PLANNED Inflation-adjusted returns
53. PLANNED Currency formatting by locale
54. PLANNED Speech read-out of results

3 · Numbered key map



Calculator #9 with key numbers shown (Hi-Contrast colours).

Key table

"Shift" = 2nd / f / SHIFT layer; "Alpha" = ALPHA layer; "3rd" = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	CPT	QUIT	·	·	op
2	ENTER	SET	·	·	op
3	↑	DEL	·	·	mod
4	↓	INS	·	·	mod
5	ON|OFF	·	·	·	clr
6	2ND	·	·	·	mod
7	CF	·	·	·	fn
8	NPV	·	·	·	fn
9	IRR	·	·	·	fn
10	→	·	·	·	clr
11	N	xP/Y	·	·	mem
12	I/Y	P/Y	·	·	mem
13	PV	AMORT	·	·	mem
14	PMT	BGN	·	·	mem
15	FV	CLR TVM	·	·	mem
16	%	K	·	·	op
17	√x	·	·	·	fn
18	x^2	·	·	·	fn
19	1/x	·	·	·	fn
20	÷	RAND	·	·	op
21	INV	HYP	·	·	mod
22	(	SIN	·	·	op
23	)	COS	·	·	op

#	Face	Shift	Alpha	3rd	Group
24	yx	TAN	·	·	op
25	×	x!	·	·	op
26	LN	ex	·	·	fn
27	7	DATA	·	·	num
28	8	STAT	·	·	num
29	9	BOND	·	·	num
30	−	nPr	·	·	op
31	STO	ROUND	·	·	mem
32	4	DEPR	·	·	num
33	5	Δ%	·	·	num
34	6	BRKEVN	·	·	num
35	+	nCr	·	·	op
36	RCL	·	·	·	mem
37	1	DATE	·	·	num
38	2	ICONV	·	·	num
39	3	PROFIT	·	·	num
40	=	ANS	·	·	op
41	CE|C	CLR WORK	·	·	clr
42	0	MEM	·	·	num
43	.	FORMAT	·	·	num
44	+|−	RESET	·	·	num

4 · Key-by-key reference

Every key on calculator #9, in number order, with every function it carries in this app.

#1 CPT `Operators & execute`

Primary: CPT — compute the TVM / worksheet variable pressed next

shifted (2nd / f / SHIFT layer) (press #6 first): QUIT — QUIT — leave the current screen / app

#2 ENTER `Operators & execute`

Primary: ENTER — store the typed value in the worksheet item

shifted (2nd / f / SHIFT layer) (press #6 first): SET — SET — toggle a worksheet setting

Keyboard: `Enter`

#3 ↑ `Shift, modes, menus & navigation`

Primary: ↑ — previous worksheet item

shifted (2nd / f / SHIFT layer) (press #6 first): DEL — DEL — delete the character at the cursor

Keyboard: `ArrowUp`

#4 ↓ Shift, modes, menus & navigation

Primary: ↓ — next worksheet item

shifted (2nd / f / SHIFT layer) (press #6 first): INS — INS — insert mode at the cursor

Keyboard: `ArrowDown`

#5 ON|OFF Clear, delete & power

Primary: ON/OFF — power toggle

#6 2ND Shift, modes, menus & navigation

Primary: 2ND — the next key uses its second function

#7 CF Functions

Primary: CF worksheet

#8 NPV Functions

Primary: NPV worksheet

#9 IRR Functions

Primary: IRR worksheet

#10 → `Clear, delete & power`

Primary: Backspace — delete the last digit typed

Keyboard: `Backspace` `Delete`

#11 **N** `Memory, variables & registers`

Primary: N — number of periods (TVM)

shifted (2nd / f / SHIFT layer) (press #6 first): xP/Y — xP/Y — multiply by payments per year

#12 **I/Y** `Memory, variables & registers`

Primary: I/Y — interest rate per year (TVM)

shifted (2nd / f / SHIFT layer) (press #6 first): P/Y — PY worksheet

#13 **PV** `Memory, variables & registers`

Primary: PV — present value

shifted (2nd / f / SHIFT layer) (press #6 first): AMORT — AMORT worksheet

#14 **PMT** `Memory, variables & registers`

Primary: PMT — periodic payment

shifted (2nd / f / SHIFT layer) (press #6 first): BGN — BGN worksheet

#15 FV Memory, variables & registers

Primary: FV — future value

shifted (2nd / f / SHIFT layer) (press #6 first): CLR TVM — CLR TVM — clear N, I/Y, PV, PMT, FV

#16 % Operators & execute

Primary: Percent %

shifted (2nd / f / SHIFT layer) (press #6 first): K — K — constant: repeat the last operation with a fixed operand

Keyboard: [%]

#17 √x Functions

Primary: Square root √

Keyboard: [r]

#18 x^2 Functions

Primary: Square x^2

Keyboard: [q]

#19 1/x Functions

Primary: Reciprocal x^{-1} (1/x)

Keyboard: [i]

#20 ÷ Operators & execute

Primary: Division ÷

shifted (2nd / f / SHIFT layer) (press #6 first): RAND — Random number 0–1

Keyboard: /

#21 INV Shift, modes, menus & navigation

Primary: INV — inverse prefix for the next function key

shifted (2nd / f / SHIFT layer) (press #6 first): HYP — Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh

#22 (Operators & execute

Primary: Opening bracket (

shifted (2nd / f / SHIFT layer) (press #6 first): SIN — Sine (uses the current angle unit)

Keyboard: (

#23) Operators & execute

Primary: Closing bracket). Closing brackets at the end of a line may be omitted

shifted (2nd / f / SHIFT layer) (press #6 first): COS — Cosine (current angle unit)

Keyboard:)

#24 yx `Operators & execute`

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #6 first): TAN — Tangent (current angle unit)

Keyboard: [^]

#25 × `Operators & execute`

Primary: Multiplication ×

shifted (2nd / f / SHIFT layer) (press #6 first): x! — Factorial n! (also on non-integers via the gamma function where supported)

Keyboard: [*] [x]

#26 LN `Functions`

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #6 first): ex — e^x — exponential

Keyboard: [n]

#27 7 `Digits & number entry`

Primary: Digit 7

shifted (2nd / f / SHIFT layer) (press #6 first): DATA — DATA worksheet

Keyboard: [7]

#28 8 Digits & number entry

Primary: Digit 8

shifted (2nd / f / SHIFT layer) (press #6 first): STAT — STAT worksheet

Keyboard: 8

#29 9 Digits & number entry

Primary: Digit 9

shifted (2nd / f / SHIFT layer) (press #6 first): BOND — BOND worksheet

Keyboard: 9

#30 – Operators & execute

Primary: Subtraction –

shifted (2nd / f / SHIFT layer) (press #6 first): nPr — Permutations nPr

Keyboard: -

#31 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

shifted (2nd / f / SHIFT layer) (press #6 first): ROUND — Round to the displayed precision

#32 4 Digits & number entry

Primary: Digit 4

shifted (2nd / f / SHIFT layer) (press #6 first): DEPR — DEPR worksheet

Keyboard: 4

#33 5 Digits & number entry

Primary: Digit 5

shifted (2nd / f / SHIFT layer) (press #6 first): Δ% — DCH worksheet

Keyboard: 5

#34 6 Digits & number entry

Primary: Digit 6

shifted (2nd / f / SHIFT layer) (press #6 first): BRKEVN — BRKEVN worksheet

Keyboard: 6

#35 + Operators & execute

Primary: Addition +

shifted (2nd / f / SHIFT layer) (press #6 first): nCr — Combinations nCr

Keyboard: +

#36 RCL Memory, variables & registers

Primary: RCL — recall a memory / variable

#37 **1** `Digits & number entry`

Primary: Digit 1

shifted (2nd / f / SHIFT layer) (press #6 first): DATE — DATE worksheet

Keyboard: 1

#38 **2** `Digits & number entry`

Primary: Digit 2

shifted (2nd / f / SHIFT layer) (press #6 first): ICONV — ICONV worksheet

Keyboard: 2

#39 **3** `Digits & number entry`

Primary: Digit 3

shifted (2nd / f / SHIFT layer) (press #6 first): PROFIT — PROFIT worksheet

Keyboard: 3

#40 **=** `Operators & execute`

Primary: = — complete the calculation (chain / AOS)

shifted (2nd / f / SHIFT layer) (press #6 first): ANS — Ans — recall the last answer into the current expression

Keyboard: =

#41 CE|C `Clear, delete & power`

Primary: CE/C — clear entry, press twice to clear the calculation

shifted (2nd / f / SHIFT layer) (press #6 first): CLR WORK — CLR WORK — clear the open worksheet

Keyboard: `Escape`

#42 0 `Digits & number entry`

Primary: Digit 0

shifted (2nd / f / SHIFT layer) (press #6 first): MEM — MEM worksheet

Keyboard: `0`

#43 . `Digits & number entry`

Primary: Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

shifted (2nd / f / SHIFT layer) (press #6 first): FORMAT — FORMAT worksheet

Keyboard: `.` `,`

#44 +|– `Digits & number entry`

Primary: Change sign (+/–) of the number being typed or the current result

shifted (2nd / f / SHIFT layer) (press #6 first): RESET — RESET — return to factory defaults

5 · Worked examples

20 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Order of operations

Calculate 12 + 3 × 4. Press Reset first.

#37 1 → #38 2 → #35 + → #22 (→ #39 3 → #25 × → #32 4 → #23) → #40 =

Display shows 24.00 **— expected 24. ✔ verified on the app.**

Multiplication is done before addition, so the calculator holds the + until the × part is finished.

Example 5.2 — Brackets first

Calculate (12 + 8) ÷ 4. Press Reset first.

#22 (→ #37 1 → #38 2 → #35 + → #28 8 → #23) → #20 ÷ → #32 4 → #40 =

Display shows 5.00 **— expected 5. ✔ verified on the app.**

Brackets force the sum to be worked out before the division.

Example 5.3 — Two brackets multiplied

Calculate (3 + 4) × (9 − 2). Press Reset first.

#22 (→ #39 3 → #35 + → #32 4 → #23) → #25 × → #22 (→ #29 9 → #30 − → #38 2 → #23) → #40 =

Display shows 49.00 **— expected 49. ✔ verified on the app.**

Both brackets are evaluated first, then multiplied.

Example 5.4 — Squares

Calculate 12^2. Press Reset first.

#37 1 → #38 2 → #18 x^2 → #40 =

Display shows 144.00 **— expected 144. ✔ verified on the app.**

x^2 squares the value — quicker and safer than typing the number twice.

Example 5.5 — Square roots

Calculate √144. Press Reset first.

#37 1 → #32 4 → #32 4 → #17 √x → #40 =

Display shows 12.00 **— expected 12. ✔ verified on the app.**

The square-root key returns the principal (positive) root.

Example 5.6 — Powers

Calculate 2^10. Press Reset first.

#38 2 → #24 y^x → #37 1 → #42 0 → #40 =

Display shows 1024.00 **— expected 1024. ✔ verified on the app.**

The power key raises the first number to the second.

Example 5.7 — Reciprocals

Calculate 1 ÷ 4 (x^{-1}). Press Reset first.

#32 4 → #19 1/*x* → #40 =

Display shows 0.25 **— expected 0.25. ✔ verified on the app.**

x^{-1} (1/x) gives the reciprocal — handy for resistors in parallel or rates.

Example 5.8 — Trigonometry (degrees)

Calculate sin 30° (Deg mode). Press Reset first.

#39 3 → #42 0 → #6 2ND → #22 SIN → #40 =

Display shows 0.50 **— expected 0.5. ✔ verified on the app.**

Trig functions use the current angle unit; these answers assume degrees (the start-up setting).

Example 5.9 — Natural logarithm

Calculate ln 1. Press Reset first.

#37 1 → **#26** LN → **#40** =

Display shows 0.00 **— expected 0. ✔ verified on the app.**

ln is the natural (base-e) logarithm.

Example 5.10 — Exponential e^x

Calculate e^1. Press Reset first.

#37 1 → **#6** 2ND → **#26** e^x → **#40** =

Display shows 2.72 **— expected 2.71828183. ✔ verified on the app.**

e^x is the inverse of ln.

Example 5.11 — Negative numbers

Calculate −5 + 12. Press Reset first.

#33 5 → **#44** +|− → **#35** + → **#37** 1 → **#38** 2 → **#40** =

Display shows 7.00 **— expected 7. ✔ verified on the app.**

Use the sign key for a negative number — not the subtraction key.

Example 5.12 — Combinations nCr

Calculate **5C2**. Press **Reset** first.

#33 5 → **#6** 2ND → **#35** *nCr* → **#38** 2 → **#40** =

Display shows 10.00 — expected 10. ✔ verified on the app.

nCr counts selections where order does not matter.

Example 5.13 — Factorials

Calculate **5!**. Press **Reset** first.

#33 5 → **#6** 2ND → **#25** *x!* → **#40** =

Display shows 120.00 — expected 120. ✔ verified on the app.

n! multiplies 1×2×…×n; 0! is 1 by definition.

Example 5.14 — Continue from the answer

Calculate **12 × 3 =** , then **Ans ÷ 5**. Press **Reset** first.

#37 1 → **#38** 2 → **#25** × → **#39** 3 → **#40** = → **#20** ÷ → **#33** 5 → **#40** =

Display shows 7.20 — expected 7.2. ✔ verified on the app.

Pressing an operator straight after a result continues from that answer (Ans).

Example 5.15 — Percent of an amount

Calculate 15 % of 80. Press Reset first.

#28 8 → #42 0 → #25 × → #37 1 → #33 5 → #16 % → #40 =

Display shows 12.00 **— expected 12. ✔ verified on the app.**

The percent key turns the second number into a percentage of the first.

Example 5.16 — Average of three numbers

Calculate (4 + 8 + 15) ÷ 3. Press Reset first.

#22 (→ #32 4 → #35 + → #28 8 → #35 + → #37 1 → #33 5 → #23) → #20 ÷ → #39 3 → #40 =

Display shows 9.00 **— expected 9. ✔ verified on the app.**

Add up the values, then divide by how many there are.

Example 5.17 — Mixed operations

Calculate 50 − 4 × 6 + 7. Press Reset first.

#33 5 → #42 0 → #30 − → #22 (→ #32 4 → #25 × → #34 6 → #23) → #35 + → #27 7 → #40 =

Display shows 33.00 **— expected 33. ✔ verified on the app.**

Only the × part is done first; the − and + are then worked left to right.

Example 5.18 — Dividing twice

Calculate 360 ÷ 4 ÷ 9. Press Reset first.

#39 3 → #34 6 → #42 0 → #20 ÷ → #32 4 → #20 ÷ → #29 9 → #40 =

Display shows 10.00 **— expected 10. ✔ verified on the app.**

Division is worked left to right: the first quotient is divided again.

Example 5.19 — Brackets inside a longer formula

Calculate ((3 + 5) × 4 − 2) ÷ 10. Press Reset first.

#22 (→ #22 (→ #39 3 → #35 + → #33 5 → #23) → #25 × → #32 4 → #30 − → #38 2 → #23) → #20 ÷ → #37 1 → #42 0 → #40 =

Display shows 3.00 **— expected 3. ✔ verified on the app.**

The innermost brackets are worked out first, then the outer brackets, then the division.

Example 5.20 — Loan payment (TVM)

Calculate Loan 100000, 6 % a year, 360 monthly payments → PMT. Press Reset first.

#39 3 → #34 6 → #42 0 → #11 N → #34 6 → #12 I/Y →
#37 1 → #42 0 → #42 0 → #42 0 → #42 0 → #42 0 →
#13 PV → #42 0 → #15 FV → #1 CPT → #14 PMT

Display shows PMT= -599.55 **— expected -599.55. ✔ verified on the app.**

Money paid out is negative: a positive loan (PV) produces a negative payment.

6 · Tricks & hacks

Hack 1: Sign convention

Money you receive is positive, money you pay is negative. A loan: PV = +100000, PMT comes out negative.

Hack 2: Clear TVM first

Always clear the TVM registers before a new problem so old values don't leak in.

Hack 3: Check BEG/END

Payments at the beginning of each period (leases, rent) need BEG mode.

Hack 4: Monthly rates

Convert yearly figures: n = years × 12, i = yearly % ÷ 12 (on models with periodic i).

Hack 5: Worksheets

Use ↑ ↓ to move through a worksheet, ENTER to store, CPT to compute.

Hack 6: Order of operations

Algebraic one-line models respect × ÷ before + −, so 2 + 3 × 4 = 14 without brackets.

Hack 7: Constant calculations

Repeat the last operation by pressing = again (where supported) — handy for repeated discounts or growth.

Hack 8: Memory

Use the memory keys to hold a subtotal while you calculate something else.

Hack 9: Sign change

Use +/− after typing the number to make it negative.

Hack 10: Statistics

Enter each data value with Σ+/DATA; then read mean and deviation with the shifted keys.

Hack 11: Hidden functions on this model

34 extra functions sit on shifted layers. Press the layer key once, then the key: for example #6 then #1 → QUIT; #6 then #2 → SET; #6 then #3 → DEL; #6 then #4 → INS.

Hack 12: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left→right, top→bottom from #1 to #44.

Hack 13: Type faster on a computer

29 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 14: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 15: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 16: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
`Enter`	#2	ENTER	ENTER — store the typed value in the worksheet item
`ArrowUp`	#3	↑	↑ — previous worksheet item
`ArrowDown`	#4	↓	↓ — next worksheet item
`Backspace` `Delete`	#10	→	Backspace — delete the last digit typed
`%`	#16	%	Percent %
`r`	#17	√x	Square root √
`q`	#18	x^2	Square x^2
`i`	#19	1/x	Reciprocal x^{-1} (1/x)
`/`	#20	÷	Division ÷
`(`	#22	(	Opening bracket (
`)`	#23	)	Closing bracket). Closing brackets at the end of a line may be omitted
`^`	#24	yx	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
`*` `x`	#25	×	Multiplication ×
`n`	#26	LN	Natural logarithm ln
`7`	#27	7	Digit 7
`8`	#28	8	Digit 8
`9`	#29	9	Digit 9
`-`	#30	−	Subtraction −
`4`	#32	4	Digit 4
`5`	#33	5	Digit 5

6	#34	6	Digit 6
+	#35	+	Addition +
1	#37	1	Digit 1
2	#38	2	Digit 2
3	#39	3	Digit 3
=	#40	=	= — complete the calculation (chain / AOS)
Escape	#41	CE|C	CE/C — clear entry, press twice to clear the calculation
0	#42	0	Digit 0
. ,	#43	.	Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Same sign on PV and FV

If PV and FV have the same sign, there is no interest rate solution — one must be negative.

Mistake 2: Leftover values

Not clearing TVM is the most common cause of wrong payments.

Mistake 3: Percent vs decimal

Enter rates as percentages (6 for 6 %), not 0.06.

Mistake 4: Chain logic

Financial models may evaluate left-to-right in chain mode: 2 + 3 × 4 = 20. Use brackets or switch to AOS.

Mistake 5: Typing +/− first

On one-line models +/− is pressed after the digits: 5 +/− gives −5.

Mistake 6: CE vs C

CE clears only the last entry; C/AC clears the whole calculation.

Mistake 7: Forgetting the angle unit

Check DEG/RAD/GRAD before trig functions.

Mistake 8: Functions after the number

One-line models apply functions to the number already on display: 9 √ = 3 (not √ 9).

Mistake 9: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 10: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (123)

Work each one on calculator #9, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 25 + 5 × 6 = _____
2. 7 + 8 × 9 = _____
3. 150 + 12 × 11 = _____
4. 3 + 14 × 15 = _____
5. 48 + 6 × 7 = _____
6. 2.5 + 4 × 1.5 = _____
7. 1000 + 25 × 4 = _____
8. 19 + 21 × 3 = _____
9. 64 + 8 × 8 = _____
10. 0.5 + 0.25 × 8 = _____
11. (45 + 15) ÷ 6 = _____
12. (9 + 7) ÷ 2 = _____
13. (100 + 60) ÷ 8 = _____
14. (33 + 17) ÷ 5 = _____
15. (7.5 + 2.5) ÷ 4 = _____
16. (120 + 80) ÷ 16 = _____

17. $(1 + 2) \div 4 =$ _____

18. $(250 + 350) \div 12 =$ _____

19. $(5 + 6) \times (10 - 3) =$ _____

20. $(12 + 3) \times (8 - 5) =$ _____

21. $(1.5 + 2.5) \times (7 - 4) =$ _____

22. $(20 + 30) \times (15 - 5) =$ _____

23. $(9 + 1) \times (100 - 1) =$ _____

24. $(11 + 13) \times (17 - 9) =$ _____

25. $25^2 =$ _____

26. $7^2 =$ _____

27. $1.5^2 =$ _____

28. $19^2 =$ _____

29. $33^2 =$ _____

30. $0.5^2 =$ _____

31. $125^2 =$ _____

32. $64^2 =$ _____

33. $\sqrt{625} =$ _____

34. $\sqrt{2} =$ _____

35. $\sqrt{50} =$ _____

36. $\sqrt{1024} =$ _____

37. √3 = _____

38. √0.25 = _____

39. √9801 = _____

40. 3^5 = _____

41. 1.05^3 = _____

42. 5^4 = _____

43. 7^3 = _____

44. 1.5^2 = _____

45. 10^6 = _____

46. 2^0.5 = _____

47. 1 ÷ 8 (x^{-1}) = _____

48. 1 ÷ 0.25 (x^{-1}) = _____

49. 1 ÷ 16 (x^{-1}) = _____

50. 1 ÷ 5 (x^{-1}) = _____

51. 1 ÷ 40 (x^{-1}) = _____

52. 1 ÷ 0.2 (x^{-1}) = _____

53. cos 60° (Deg mode) = _____

54. tan 45° (Deg mode) = _____

55. sin 90° (Deg mode) = _____

56. cos 0° (Deg mode) = _____

57. ln 10 = _____

58. ln 2 = _____

59. e^2 = _____

60. −20 + 7 = _____

61. −3 + 3 = _____

62. −15.5 + 4 = _____

63. −100 + 250 = _____

64. −8 + 1 = _____

65. 10C3 = _____

66. 6C6 = _____

67. 6! = _____

68. 0! = _____

69. 100 × 4 = , then Ans ÷ 2 = _____

70. 15 × 6 = , then Ans ÷ 9 = _____

71. 7.5 × 8 = , then Ans ÷ 3 = _____

72. 250 × 12 = , then Ans ÷ 100 = _____

73. 8 % of 250 = _____

74. 25 % of 360 = _____

75. 12.5 % of 40 = _____

76. 50 % of 75 = _____

77. 10 % of 450 = _____

78. 20 % of 125 = _____

79. 5 % of 640 = _____

80. 75 % of 48 = _____

81. 30 % of 90 = _____

82. (12 + 15 + 18) ÷ 3 = _____

83. (70 + 85 + 91) ÷ 3 = _____

84. (2.5 + 3.5 + 6) ÷ 3 = _____

85. (100 + 200 + 600) ÷ 3 = _____

86. 100 − 3 × 12 + 9 = _____

87. 18 − 2 × 5 + 40 = _____

88. 7.5 − 1.5 × 3 + 2 = _____

89. 64 − 8 × 4 + 10 = _____

90. 300 − 25 × 4 + 55 = _____

91. 9 − 3 × 3 + 1 = _____

92. 45 − 5 × 5 + 15 = _____

93. 1000 ÷ 8 ÷ 5 = _____

94. 96 ÷ 2 ÷ 3 = _____

95. 7200 ÷ 60 ÷ 60 = _____

96. 500 ÷ 4 ÷ 25 = _____

97. 81 ÷ 3 ÷ 3 = _____

98. 1.5 ÷ 0.5 ÷ 2 = _____

99. ((10 + 2) × 3 − 6) ÷ 6 = _____

100. ((1.5 + 2.5) × 6 − 4) ÷ 5 = _____

101. ((20 + 30) × 2 − 25) ÷ 3 = _____

102. ((7 + 8) × 9 − 15) ÷ 12 = _____

103. Loan 25000, 4.5 % a year, 60 monthly payments → PMT = _____

104. 12 + 3 × 4 = _____

105. (12 + 8) ÷ 4 = _____

106. (3 + 4) × (9 − 2) = _____

107. 12^2 = _____

108. $\sqrt{144}$ = _____

109. 2^10 = _____

110. 1 ÷ 4 (x^{-1}) = _____

111. sin 30° (Deg mode) = _____

112. ln 1 = _____

113. e^1 = _____

114. −5 + 12 = _____

115. 5C2 = _____

116. 5! = _____

117. 12 × 3 = , then Ans ÷ 5 = _____

118. 15 % of 80 = _____

119. (4 + 8 + 15) ÷ 3 = _____

120. 50 − 4 × 6 + 7 = _____

121. 360 ÷ 4 ÷ 9 = _____

122. ((3 + 5) × 4 − 2) ÷ 10 = _____

123. Loan 100000, 6 % a year, 360 monthly payments → PMT = _____

Set B — Key-finding drills (68)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “CPT” (CPT)? _____
2. Which key is “ENTER” (ENTER)? _____
3. Which key is “↑” (↑)? _____
4. Which key is “↓” (↓)? _____
5. Which key is “ON|OFF” (ON/OFF)? _____
6. Which key is “2ND” (2ND)? _____
7. Which key is “CF” (CF worksheet)? _____
8. Which key is “NPV” (NPV worksheet)? _____
9. Which key is “IRR” (IRR worksheet)? _____
10. Which key is “ → ” (Backspace)? _____
11. Which key is “N” (N)? _____

12. Which key is “I/Y” (I/Y)? _____

13. Which key is “PV” (PV)? _____

14. Which key is “PMT” (PMT)? _____

15. Which key is “FV” (FV)? _____

16. Which key is “%” (Percent %)? _____

17. Which key is “√ x” (Square root √)? _____

18. Which key is “x^2” (Square x^2)? _____

19. Which key is “1/x” (Reciprocal x^{-1})? _____

20. Which key is “÷” (Division ÷)? _____

21. Which key is “INV” (INV)? _____

22. Which key is “(” (Opening bracket)? _____

23. Which key is “)” (Closing bracket). Closing brackets at the end of a line may be omitted)? _____

24. Which key is “yx” (Power x^y / ^)? _____

25. Which key is “×” (Multiplication ×)? _____

26. Which key is “LN” (Natural logarithm ln)? _____

27. Which key is “–” (Subtraction –)? _____

28. Which key is “STO” (STO)? _____

29. Which key is “+” (Addition +)? _____

30. Which key is “RCL” (RCL)? _____

31. Which key is “=” (=)? _____

32. Which key is “CE|C” (CE/C)? _____

33. Which key is “.” (Decimal point. Pressing it first gives “0.”; a second point in the same number is ignored)? _____

34. Which key is “+|–” (Change sign)? _____

35. How do you reach “QUIT”? _____

36. How do you reach “SET”? _____

37. How do you reach “DEL”? _____

38. How do you reach “INS”? _____

39. How do you reach “xP/Y”? _____

40. How do you reach “P/Y”? _____

41. How do you reach “AMORT”? _____

42. How do you reach “BGN”? _____

43. How do you reach “CLR TVM”? _____

44. How do you reach “K”? _____

45. How do you reach “RAND”? _____

46. How do you reach “HYP”? _____

47. How do you reach “SIN”? _____

48. How do you reach “COS”? _____

49. How do you reach “TAN”? _____

50. How do you reach “x!”? _____

51. How do you reach “ex”? _____

52. How do you reach “DATA”? _____

53. How do you reach “STAT”? _____

54. How do you reach “BOND”? _____

55. How do you reach “nPr”? _____

56. How do you reach “ROUND”? _____

57. How do you reach “DEPR”? _____

58. How do you reach “Δ%”? _____

59. How do you reach “BRKEVN”? _____

60. How do you reach “nCr”? _____

61. How do you reach “DATE”? _____

62. How do you reach “ICONV”? _____

63. How do you reach “PROFIT”? _____

64. How do you reach “ANS”? _____

65. How do you reach “CLR WORK”? _____

66. How do you reach “MEM”? _____

67. How do you reach “FORMAT”? _____

68. How do you reach “RESET”? _____

Appendix A — Drill answers

Set A

1. 55 (display `55.00`)

`#38` 2 → `#33` 5 → `#35` + → `#22` (→ `#33` 5 → `#25` × → `#34` 6 → `#23`) → `#40` =

2. 79 (display `79.00`)

`#27` 7 → `#35` + → `#22` (→ `#28` 8 → `#25` × → `#29` 9 → `#23`) → `#40` =

3. 282 (display `282.00`)

`#37` 1 → `#33` 5 → `#42` 0 → `#35` + → `#22` (→ `#37` 1 → `#38` 2 → `#25` × → `#37` 1 → `#37` 1 → `#23`) → `#40` =

4. 213 (display `213.00`)

`#39` 3 → `#35` + → `#22` (→ `#37` 1 → `#32` 4 → `#25` × → `#37` 1 → `#33` 5 → `#23`) → `#40` =

5. 90 (display `90.00`)

`#32` 4 → `#28` 8 → `#35` + → `#22` (→ `#34` 6 → `#25` × → `#27` 7 → `#23`) → `#40` =

6. 8.5 (display `8.50`)

`#38` 2 → `#43` . → `#33` 5 → `#35` + → `#22` (→ `#32` 4 → `#25` × → `#37` 1 → `#43` . → `#33` 5 → `#23`) → `#40` =

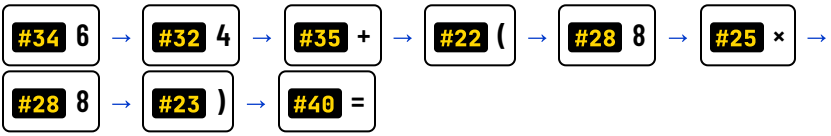
7. 1100 (display `1100.00`)



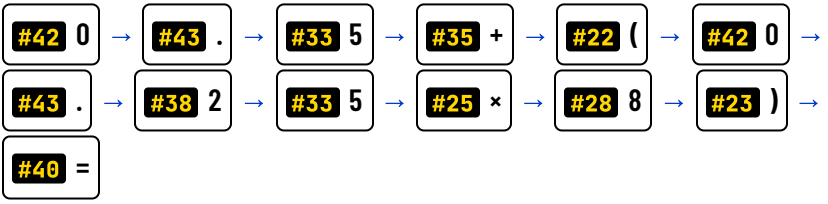
8. 82 (display `82.00`)



9. 128 (display `128.00`)



10. 2.5 (display `2.50`)



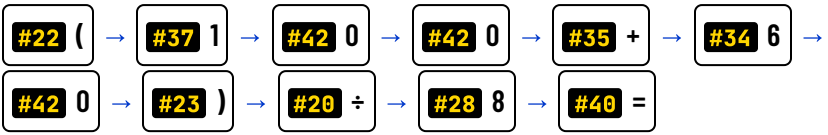
11. 10 (display `10.00`)



12. 8 (display `8.00`)



13. 20 (display 20.00)



14. 10 (display 10.00)



15. 2.5 (display 2.50)



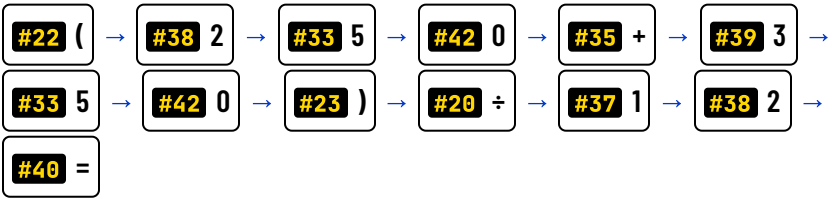
16. 12.5 (display 12.50)



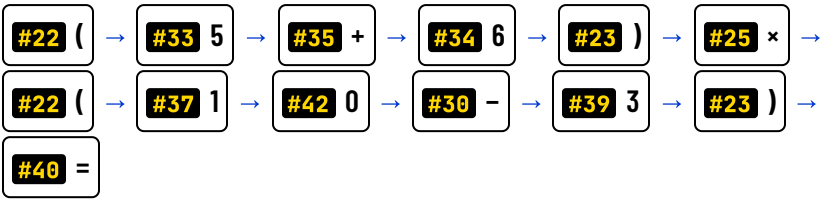
17. 0.75 (display 0.75)



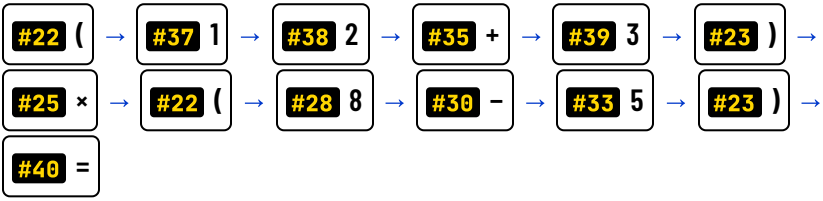
18. 50 (display 50.00)



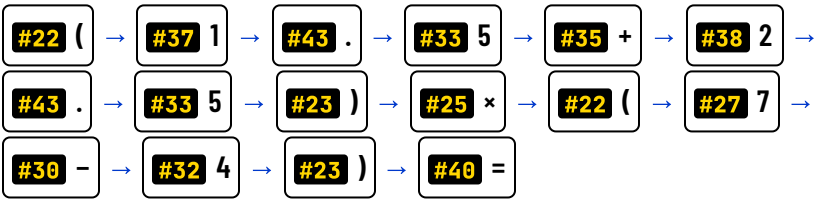
19. 77 (display `77.00`)



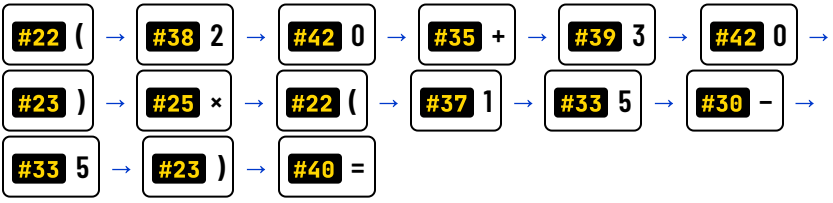
20. 45 (display `45.00`)



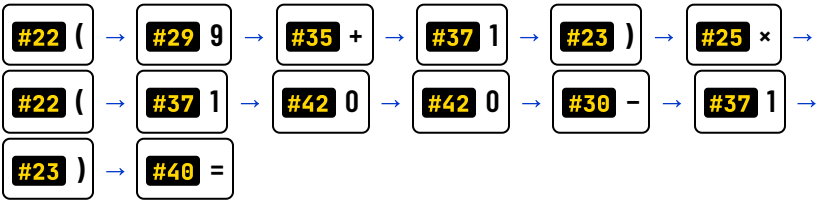
21. 12 (display `12.00`)



22. 500 (display `500.00`)



23. 990 (display `990.00`)



24. 192 (display 192.00)

#22 (→ #37 1 → #37 1 → #35 + → #37 1 → #39 3 → #23) → #25 × → #22 (→ #37 1 → #27 7 → #30 − → #29 9 → #23) → #40 =

25. 625 (display 625.00)

#38 2 → #33 5 → #18 x^2 → #40 =

26. 49 (display 49.00)

#27 7 → #18 x^2 → #40 =

27. 2.25 (display 2.25)

#37 1 → #43 . → #33 5 → #18 x^2 → #40 =

28. 361 (display 361.00)

#37 1 → #29 9 → #18 x^2 → #40 =

29. 1089 (display 1089.00)

#39 3 → #39 3 → #18 x^2 → #40 =

30. 0.25 (display 0.25)

#42 0 → #43 . → #33 5 → #18 x^2 → #40 =

31. 15625 (display 15625.00)

#37 1 → #38 2 → #33 5 → #18 x^2 → #40 =

32. 4096 (display 4096.00)

#34 6 → #32 4 → #18 x^2 → #40 =

33. 25 (display `25.00`)

`#34` 6 → `#38` 2 → `#33` 5 → `#17` √x → `#40` =

34. 1.41421356 (display `1.41`)

`#38` 2 → `#17` √x → `#40` =

35. 7.07106781 (display `7.07`)

`#33` 5 → `#42` 0 → `#17` √x → `#40` =

36. 32 (display `32.00`)

`#37` 1 → `#42` 0 → `#38` 2 → `#32` 4 → `#17` √x → `#40` =

37. 1.73205081 (display `1.73`)

`#39` 3 → `#17` √x → `#40` =

38. 0.5 (display `0.50`)

`#42` 0 → `#43` . → `#38` 2 → `#33` 5 → `#17` √x → `#40` =

39. 99 (display `99.00`)

`#29` 9 → `#28` 8 → `#42` 0 → `#37` 1 → `#17` √x → `#40` =

40. 243 (display `243.00`)

`#39` 3 → `#24` y^x → `#33` 5 → `#40` =

41. 1.157625 (display `1.16`)

`#37` 1 → `#43` . → `#42` 0 → `#33` 5 → `#24` y^x → `#39` 3 → `#40` =

42. 625 (display `625.00`)

`#33` 5 → `#24` y^x → `#32` 4 → `#40` =

43. 343 (display `343.00`)

`#27` 7 → `#24` y^x → `#39` 3 → `#40` =

44. 2.25 (display `2.25`)

`#37` 1 → `#43` . → `#33` 5 → `#24` y^x → `#38` 2 → `#40` =

45. 1000000.000 (display `1000000.00`)

`#37` 1 → `#42` 0 → `#24` y^x → `#34` 6 → `#40` =

46. 1.41421356 (display `1.41`)

`#38` 2 → `#24` y^x → `#42` 0 → `#43` . → `#33` 5 → `#40` =

47. 0.125 (display `0.13`)

`#28` 8 → `#19` $1/x$ → `#40` =

48. 4 (display `4.00`)

`#42` 0 → `#43` . → `#38` 2 → `#33` 5 → `#19` $1/x$ → `#40` =

49. 0.0625 (display `0.06`)

`#37` 1 → `#34` 6 → `#19` $1/x$ → `#40` =

50. 0.2 (display `0.20`)

`#33` 5 → `#19` $1/x$ → `#40` =

51. 0.025 (display `0.03`)

`#32` 4 → `#42` 0 → `#19` $1/x$ → `#40` =

52. 5 (display `5.00`)

`#42` 0 → `#43` . → `#38` 2 → `#19` 1/*x* → `#40` =

53. 0.5 (display `0.50`)

`#34` 6 → `#42` 0 → `#6` 2ND → `#23` COS → `#40` =

54. 1 (display `1.00`)

`#32` 4 → `#33` 5 → `#6` 2ND → `#24` TAN → `#40` =

55. 1 (display `1.00`)

`#29` 9 → `#42` 0 → `#6` 2ND → `#22` SIN → `#40` =

56. 1 (display `1.00`)

`#42` 0 → `#6` 2ND → `#23` COS → `#40` =

57. 2.30258509 (display `2.30`)

`#37` 1 → `#42` 0 → `#26` LN → `#40` =

58. 0.69314718 (display `0.69`)

`#38` 2 → `#26` LN → `#40` =

59. 7.3890561 (display `7.39`)

`#38` 2 → `#6` 2ND → `#26` e^x → `#40` =

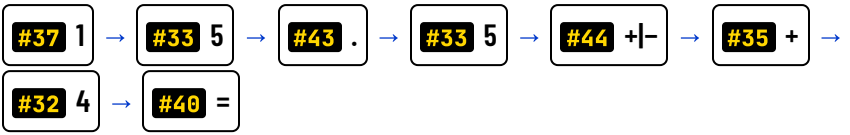
60. -13 (display `-13.00`)

`#38` 2 → `#42` 0 → `#44` +|– → `#35` + → `#27` 7 → `#40` =

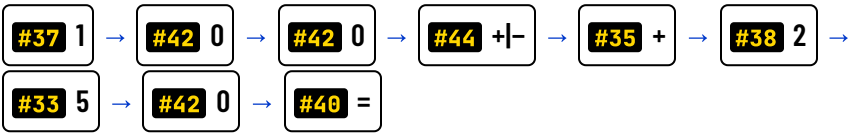
61. 0 (display `0.00`)

`#39` 3 → `#44` +|– → `#35` + → `#39` 3 → `#40` =

62. -11.5 (display `-11.50`)



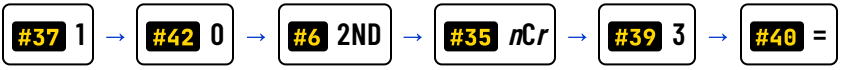
63. 150 (display `150.00`)



64. -7 (display `-7.00`)



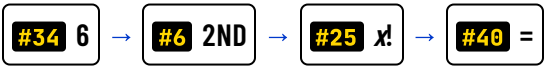
65. 120 (display `120.00`)



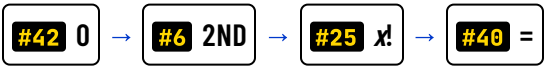
66. 1 (display `1.00`)



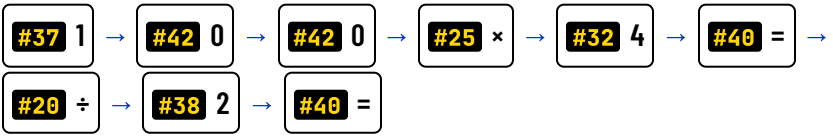
67. 720 (display `720.00`)



68. 1 (display `1.00`)



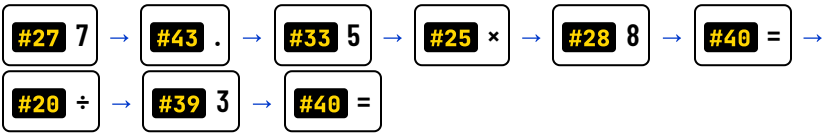
69. 200 (display `200.00`)



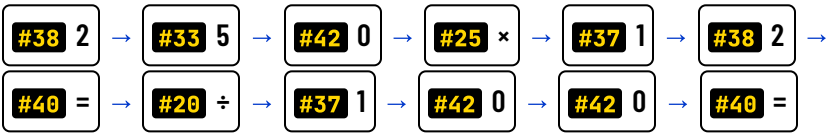
70. 10 (display 10.00)



71. 20 (display 20.00)



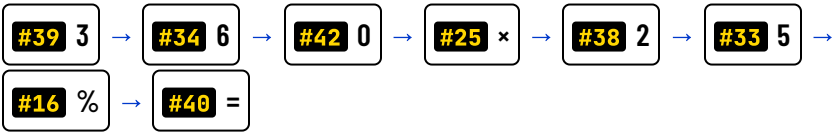
72. 30 (display 30.00)



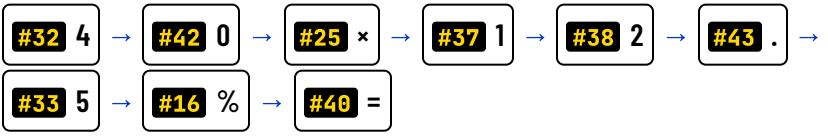
73. 20 (display 20.00)



74. 90 (display 90.00)



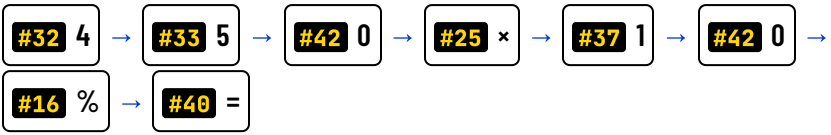
75. 5 (display 5.00)



76. 37.5 (display 37.50)



77. 45 (display 45.00)



78. 25 (display 25.00)



79. 32 (display 32.00)



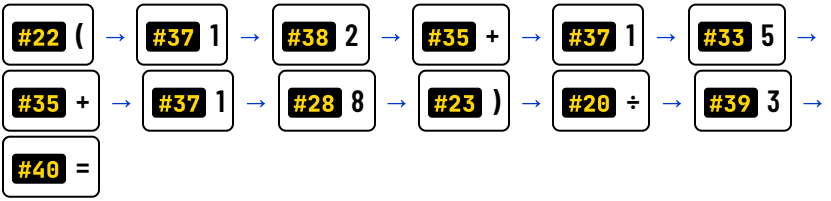
80. 36 (display 36.00)



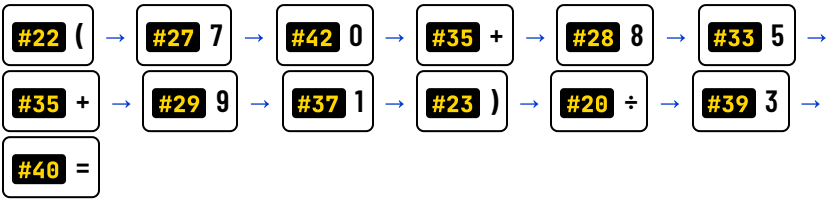
81. 27 (display 27.00)



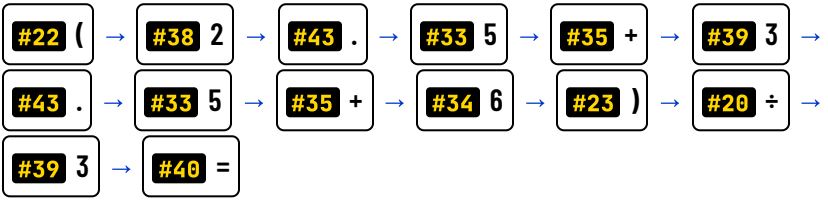
82. 15 (display 15.00)



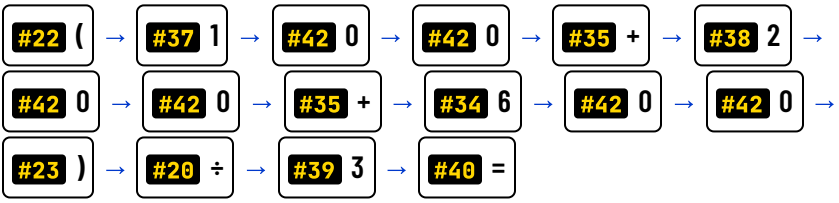
83. 82 (display 82.00)



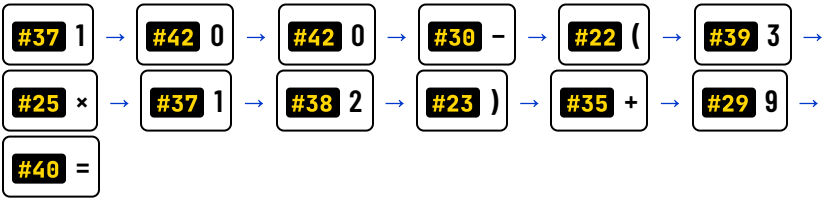
84. 4 (display 4.00)



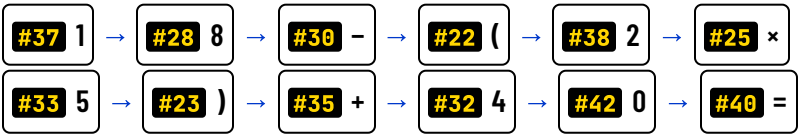
85. 300 (display 300.00)



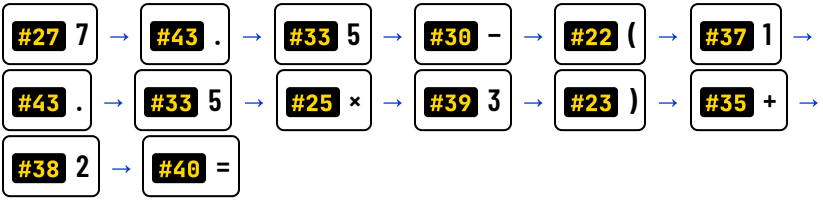
86. 73 (display 73.00)



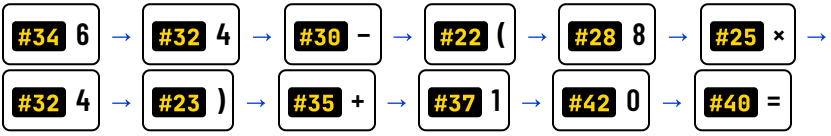
87. 48 (display 48.00)



88. 5 (display 5.00)



89. 42 (display 42.00)



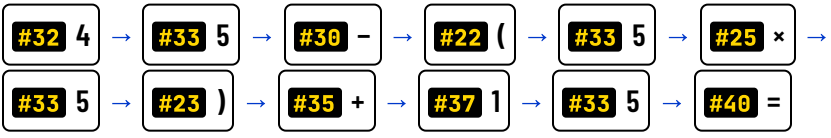
90. 255 (display 255.00)



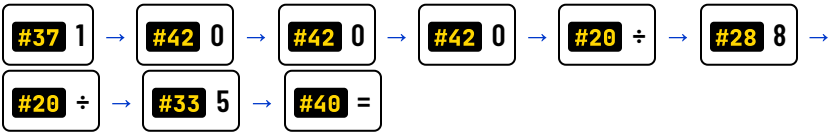
91. 1 (display 1.00)



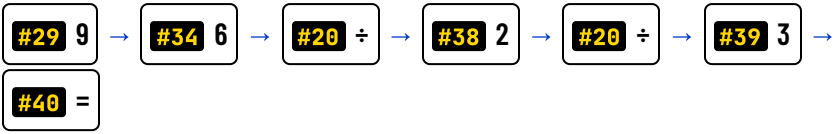
92. 35 (display 35.00)



93. 25 (display 25.00)



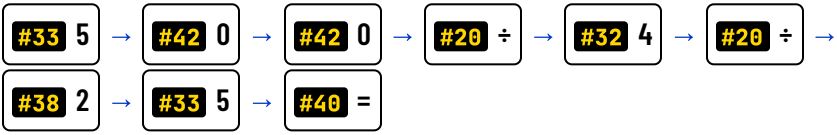
94. 16 (display `16.00`)



95. 2 (display `2.00`)



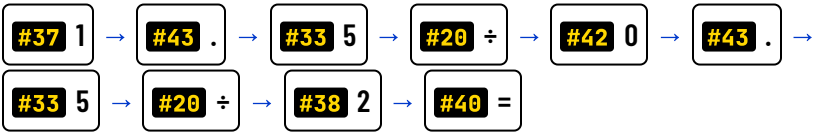
96. 5 (display `5.00`)



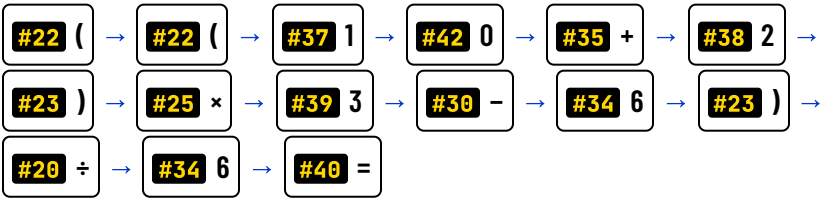
97. 9 (display `9.00`)



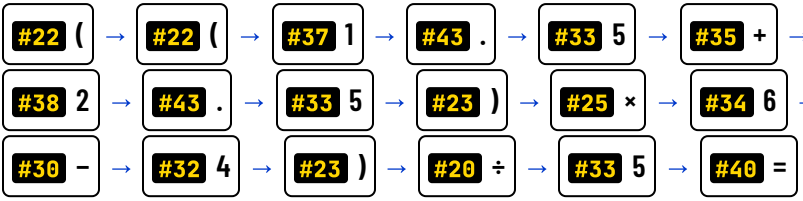
98. 1.5 (display `1.50`)



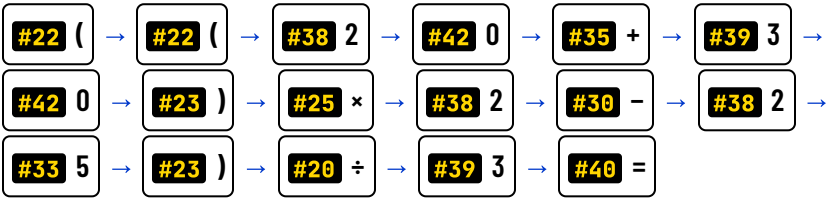
99. 5 (display `5.00`)



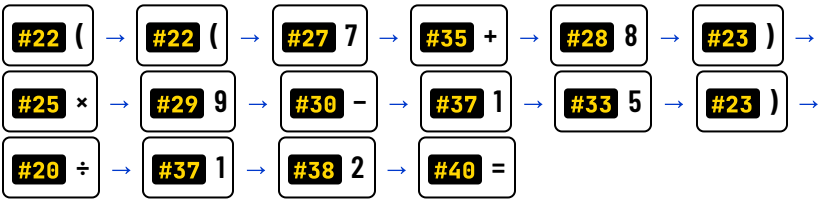
100. 4 (display `4.00`)



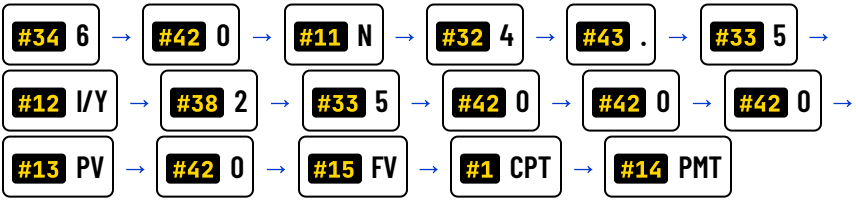
101. 25 (display `25.00`)



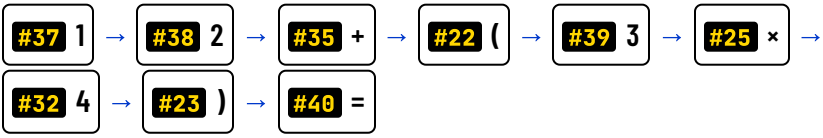
102. 10 (display `10.00`)



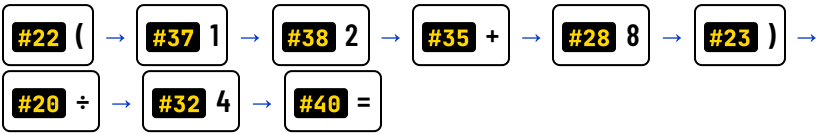
103. -466.08 (display `PMT= -466.08`)



104. 24 (display `24.00`)



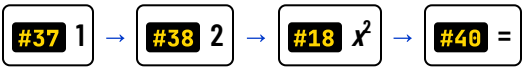
105. 5 (display 5.00)



106. 49 (display 49.00)



107. 144 (display 144.00)



108. 12 (display 12.00)



109. 1024 (display 1024.00)



110. 0.25 (display 0.25)



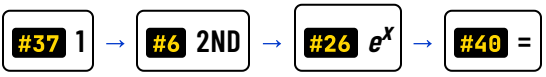
111. 0.5 (display 0.50)



112. 0 (display 0.00)



113. 2.71828183 (display 2.72)



114. 7 (display 7.00)

#33 5 → #44 +/− → #35 + → #37 1 → #38 2 → #40 =

115. 10 (display 10.00)

#33 5 → #6 2ND → #35 *nCr* → #38 2 → #40 =

116. 120 (display 120.00)

#33 5 → #6 2ND → #25 *x!* → #40 =

117. 7.2 (display 7.20)

#37 1 → #38 2 → #25 × → #39 3 → #40 = → #20 ÷ → #33 5 → #40 =

118. 12 (display 12.00)

#28 8 → #42 0 → #25 × → #37 1 → #33 5 → #16 % → #40 =

119. 9 (display 9.00)

#22 (→ #32 4 → #35 + → #28 8 → #35 + → #37 1 → #33 5 → #23) → #20 ÷ → #39 3 → #40 =

120. 33 (display 33.00)

#33 5 → #42 0 → #30 − → #22 (→ #32 4 → #25 × → #34 6 → #23) → #35 + → #27 7 → #40 =

121. 10 (display 10.00)

#39 3 → #34 6 → #42 0 → #20 ÷ → #32 4 → #20 ÷ → #29 9 → #40 =

122. 3 (display 3.00)

#22 (→ #22 (→ #39 3 → #35 + → #33 5 → #23) →
#25 × → #32 4 → #30 − → #38 2 → #23) → #20 ÷ →
#37 1 → #42 0 → #40 =

123. -599.55 (display PMT= -599.55)

#39 3 → #34 6 → #42 0 → #11 N → #34 6 → #12 I/Y →
#37 1 → #42 0 → #42 0 → #42 0 → #42 0 → #42 0 →
#13 PV → #42 0 → #15 FV → #1 CPT → #14 PMT

Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #30
28. #31
29. #35
30. #36
31. #40
32. #41
33. #43
34. #44
35. #6 2ND then #1 CPT
36. #6 2ND then #2 ENTER
37. #6 2ND then #3 ↑
38. #6 2ND then #4 ↓
39. #6 2ND then #11 N
40. #6 2ND then #12 I/Y
41. #6 2ND then #13 PV
42. #6 2ND then #14 PMT
43. #6 2ND then #15 FV
44. #6 2ND then #16 %
45. #6 2ND then #20 ÷
46. #6 2ND then #21 INV

47. #6 2ND then #22 (
48. #6 2ND then #23)
49. #6 2ND then #24 yx
50. #6 2ND then #25 ×
51. #6 2ND then #26 LN
52. #6 2ND then #27 7
53. #6 2ND then #28 8
54. #6 2ND then #29 9
55. #6 2ND then #30 -
56. #6 2ND then #31 STO
57. #6 2ND then #32 4
58. #6 2ND then #33 5
59. #6 2ND then #34 6
60. #6 2ND then #35 +
61. #6 2ND then #37 1
62. #6 2ND then #38 2
63. #6 2ND then #39 3
64. #6 2ND then #40 =
65. #6 2ND then #41 CE|C
66. #6 2ND then #42 0
67. #6 2ND then #43 .
68. #6 2ND then #44 +|-

Appendix B — Function finder (A–Z)

Function	Label	Keys
↑ — previous worksheet item	↑	#3
↓ — next worksheet item	↓	#4
= — complete the calculation (chain / AOS)	=	#40
2ND — the next key uses its second function	2ND	#6
Addition +	+	#35
AMORT worksheet	AMORT	#6 → #13
Ans — recall the last answer into the current expression	ANS	#6 → #40
Backspace — delete the last digit typed	→	#10
BGN worksheet	BGN	#6 → #14
BOND worksheet	BOND	#6 → #29
BRKEVN worksheet	BRKEVN	#6 → #34
CE/C — clear entry, press twice to clear the calculation	CE|C	#41
CF worksheet	CF	#7
Change sign (+/−) of the number being typed or the current result	+|–	#44
Closing bracket). Closing brackets at the end of a line may be omitted	)	#23
CLR TVM — clear N, I/Y, PV, PMT, FV	CLR TVM	#6 → #15
CLR WORK — clear the open worksheet	CLR WORK	#6 → #41
Combinations nCr	nCr	#6 → #35
Cosine (current angle unit)	COS	#6 → #23

CPT — compute the TVM / worksheet variable pressed next	CPT	#1
DATA worksheet	DATA	#6 → #27
DATE worksheet	DATE	#6 → #37
DCH worksheet	Δ%	#6 → #33
Decimal point. Pressing it first gives "0."; a second point in the same number is ignored	.	#43
DEL — delete the character at the cursor	DEL	#6 → #3
DEPR worksheet	DEPR	#6 → #32
Digit 0	0	#42
Digit 1	1	#37
Digit 2	2	#38
Digit 3	3	#39
Digit 4	4	#32
Digit 5	5	#33
Digit 6	6	#34
Digit 7	7	#27
Digit 8	8	#28
Digit 9	9	#29
Division ÷	÷	#20
ENTER — store the typed value in the worksheet item	ENTER	#2
e^x — exponential	ex	#6 → #26
Factorial n! (also on non-integers via the gamma function where supported)	x!	#6 → #25
FORMAT worksheet	FORMAT	#6 → #43
FV — future value	FV	#15

Hyperbolic prefix — the next sin/cos/tan becomes sinh/cosh/tanh	HYP	#6 → #21
I/Y — interest rate per year (TVM)	I/Y	#12
ICONV worksheet	ICONV	#6 → #38
INS — insert mode at the cursor	INS	#6 → #4
INV — inverse prefix for the next function key	INV	#21
IRR worksheet	IRR	#9
K — constant: repeat the last operation with a fixed operand	K	#6 → #16
MEM worksheet	MEM	#6 → #42
Multiplication ×	×	#25
N — number of periods (TVM)	N	#11
Natural logarithm ln	LN	#26
NPV worksheet	NPV	#8
ON/OFF — power toggle	ON|OFF	#5
Opening bracket (	(	#22
Percent %	%	#16
Permutations nPr	nPr	#6 → #30
PMT — periodic payment	PMT	#14
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	yx	#24
PROFIT worksheet	PROFIT	#6 → #39
PV — present value	PV	#13
P/Y worksheet	P/Y	#6 → #12
QUIT — leave the current screen / app	QUIT	#6 → #1

Random number 0–1	RAND	#6 → #20
RCL — recall a memory / variable	RCL	#36
Reciprocal x^{-1} (1/x)	1/x	#19
RESET — return to factory defaults	RESET	#6 → #44
Round to the displayed precision	ROUND	#6 → #31
SET — toggle a worksheet setting	SET	#6 → #2
Sine (uses the current angle unit)	SIN	#6 → #22
Square root √	√x	#17
Square x^2	x^2	#18
STAT worksheet	STAT	#6 → #28
STO — store the value into a memory / variable (press the target next)	STO	#31
Subtraction –	–	#30
Tangent (current angle unit)	TAN	#6 → #24
xP/Y — multiply by payments per year	xP/Y	#6 → #11

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

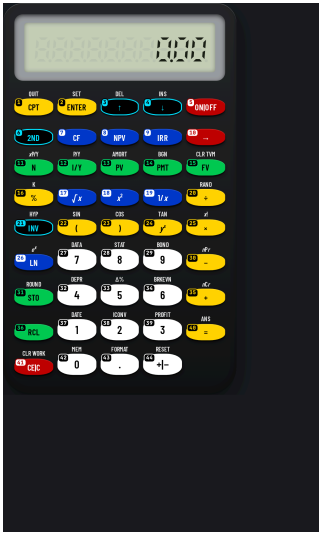
Financial algebraic model. TVM: key in values then N, I/Y, PV, PMT, FV and press CPT + the unknown. 2ND P/Y, BGN, AMORT, CF/NPV/IRR (cash-flow list with frequencies), DATA/STAT, BOND, DEPR (straight-line only), Δ%, BRKEVN, PROFIT, ICONV, DATE, MEM, FORMAT worksheets; ↑/↓ move between worksheet variables, ENTER stores, CPT computes.

Planned (not yet working)

- **MIRR / NFV / payback**
- **SYD, DB, DDB depreciation**
- **Bond yield to call**
- **Loan-comparison side-by-side**
- **Amortisation table export (CSV)**
- **Inflation-adjusted returns**
- **Currency formatting by locale**
- **Speech read-out of results**

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 CPT ⇧ QUIT	#2 ENTER ⇧ SET	#3 ↑ ⇧ DEL
#4 ↓ ⇧ INS	#5 ON|OFF	#6 2ND
#7 CF	#8 NPV	#9 IRR
#10 →	#11 N ⇧ xP/Y	#12 I/Y ⇧ P/Y
#13 PV ⇧ AMORT	#14 PMT ⇧ BGN	#15 FV ⇧ CLR TVM
#16 % ⇧ K	#17 √x	#18 x^2
#19 1/x	#20 ÷ ⇧ RAND	#21 INV ⇧ HYP
#22 (⇧ SIN	#23) ⇧ COS	#24 yx ⇧ TAN
#25 × ⇧ x!	#26 LN ⇧ ex	#27 7 ⇧ DATA
#28 8 ⇧ STAT	#29 9 ⇧ BOND	#30 − ⇧ nPr
#31 STO ⇧ ROUND	#32 4 ⇧ DEPR	#33 5 ⇧ Δ%
#34 6 ⇧ BRKEVN	#35 + ⇧ nCr	#36 RCL
#37 1 ⇧ DATE	#38 2 ⇧ ICONV	#39 3 ⇧ PROFIT
#40 = ⇧ ANS	#41 CE|C ⇧ CLR WORK	#42 0 ⇧ MEM
#43 . ⇧ FORMAT	#44 +|− ⇧ RESET	

U-45 CALCULATORS · Calculator #9 · cut out and keep